Tutorial 4 Solution

Problem 1

Determine the (first-order) Taylor approximation of

$$f : \mathbb{R}^2_{++} \to \mathbb{R}, (x, y) \mapsto \ln(3x + 2y)$$

around the point $\boldsymbol{a} = (2, 1)$.

Solution: Since

$$\nabla f(x, y) = \left(\frac{\partial f(x, y)}{\partial x}, \frac{\partial f(x, y)}{\partial y} \right) = \left(\frac{3}{3x + 2y}, \frac{2}{3x + 2y} \right),$$

the first-order Taylor approximation around around the point $\boldsymbol{a} = (2, 1)$ equals

$$\begin{aligned} f(\boldsymbol{a}) + \nabla f(\boldsymbol{a}) \cdot (\boldsymbol{x} - \boldsymbol{a}) &= \ln(8) + \left(\frac{3}{8}, \frac{2}{8} \right) \cdot (x - 2, y - 1) \\ &= \ln(8) + \frac{3}{8}(x - 2) + \frac{2}{8}(y - 1) \\ &= \ln(8) + \frac{3}{8}x + \frac{2}{8}y - 1 \end{aligned}$$

Problem 2

Suppose $x(z) = z^2 + 1$, $y(z) = e^z$, and

$$f(x, y) = \frac{3x^2}{y}.$$

Let $F(z) \equiv f(x(z), y(z))$; calculate $F'(z)$ at $z = 0$.

Solution: We use the (generalized) chain rule to derive $F'(z)$:

$$F'(z) = \frac{\partial f(x(z), y(z))}{\partial x} x'(z) + \frac{\partial f(x(z), y(z))}{\partial y} y'(z).$$

We have

$$\frac{\partial f(x, y)}{\partial x} = \frac{6x}{y}, \quad \text{and} \quad \frac{\partial f(x, y)}{\partial y} = -\frac{3x^2}{y^2}.$$

Evaluating at $x(z) = z^2 + 1$ and $y(z) = e^z$:

$$\frac{\partial f(x(z), y(z))}{\partial x} = \frac{6(z^2+1)}{e^z}, \quad \text{and} \quad \frac{\partial f(x(z), y(z))}{\partial y} = -\frac{3(z^2+1)^2}{e^{2z}}.$$

Furthermore, $x'(z) = 2z$, and $y'(z) = e^z$. Therefore,

$$F'(z) = \frac{12z(z^2+1)}{e^z} - \frac{3(z^2+1)^2}{e^z} = \frac{12z(z^2+1) - 3(z^2+1)^2}{e^z}.$$

Evaluating at $z = 0$ gets us

$$F'(0) = \frac{12(0)(0^2+1) - 3(0^2+1)^2}{e^0} = \frac{-3}{1} = -3.$$

Problem 3

Suppose that it follows from the implicit function theorem that, in a neighborhood of (x_0, y_0, z_0), z is implicitly defined by the equation $ze^x + e^y - ye^z = 0$ as a function of x and y. Compute the partial derivatives $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ at the point (x_0, y_0).

Solution: The partial derivatives of $f : \mathbb{R}^3 \to \mathbb{R}, (x, y, z) \mapsto ze^x + e^y - ye^z$ at (x_0, y_0, z_0) are:

$$\begin{aligned}
\frac{\partial f(x_0, y_0, z_0)}{\partial x} &= z_0 e^{x_0} \\
\frac{\partial f(x_0, y_0, z_0)}{\partial y} &= e^{y_0} - e^{z_0} \\
\frac{\partial f(x_0, y_0, z_0)}{\partial z} &= e^{x_0} - y_0 e^{z_0}
\end{aligned}$$

We must have $\dfrac{\partial f(x_0, y_0, z_0)}{\partial z} \neq 0$, otherwise, the implicit function theorem would not apply. Using the implicit function theorem, we obtain:

$$\begin{aligned}
\frac{\partial z(x_0, y_0)}{\partial x} &= -\frac{\frac{\partial f(x_0, y_0, z_0)}{\partial x}}{\frac{\partial f(x_0, y_0, z_0)}{\partial z}} = -\frac{z_0 e^{x_0}}{e^{x_0} - y_0 e^{z_0}} \\
\frac{\partial z(x_0, y_0)}{\partial y} &= -\frac{\frac{\partial f(x_0, y_0, z_0)}{\partial y}}{\frac{\partial f(x_0, y_0, z_0)}{\partial z}} = -\frac{e^{y_0} - e^{z_0}}{e^{x_0} - y_0 e^{z_0}} = \frac{e^{z_0} - e^{y_0}}{e^{x_0} - y_0 e^{z_0}}
\end{aligned}$$

Problem 4

Obtain an equation of the tangent line to the level curve of $f : \mathbb{R}^2 \to \mathbb{R}$ at the point

1. $(x_0, y_0) = (1, 1)$ if $f(x, y) = xe^{xy}$
2. $(x_0, y_0) = (-1, 2)$ if $f(x, y) = y - x^2$

3. **(Extra Practice Problem)** $(x_0, y_0) = (1, 2)$ if $f(x, y) = yx$

Solution:

1. Since $f(1, 1) = e$, the level curve of f through $(x_0, y_0) = (1, 1)$ is $L = \{(x, y) \in \mathbb{R}^2 : xe^{xy} = e\}$. From

$$\frac{\partial f(x,y)}{\partial x} = e^{xy} + xye^{xy} = e^{xy}(1 + xy)$$
$$\frac{\partial f(x,y)}{\partial y} = x^2 e^{xy}$$

we can conclude that f is continuously differentiable (as both its partial derivatives are continuous), and that $\frac{\partial f(1,1)}{\partial y} = e \neq 0$. Hence by the implicit function theorem, a continuously differentiable function $y = g(x)$ whose graph describes L in a vicinity of $(1, 1)$ exists, and

$$g'(1) = -\frac{\frac{\partial f(1,1)}{\partial x}}{\frac{\partial f(1,1)}{\partial y}} = -\frac{e^{1\cdot 1}(1 + 1 \cdot 1)}{1^2 e^{1\cdot 1}} = -2$$

Therefore, the equation of the tangent line g to the level curve L at $(1, 1)$ is

$$h(x) = y_0 + g'(x_0)(x - x_0) = 1 - 2(x - 1) = -2x + 3$$

2. Similarly, in this case, from

$$\frac{\partial f(x,y)}{\partial x} = -2x$$
$$\frac{\partial f(x,y)}{\partial y} = 1$$

it follows that f is continuously differentiable, and that $\frac{\partial f(-1,2)}{\partial y} = 1 \neq 0$. Hence by the implicit function theorem, a continuously differentiable function $y = g(x)$ whose graph describes L in a vicinity of $(-1, 2)$ exists, and

$$g'(-1) = -\frac{\frac{\partial f(-1,2)}{\partial x}}{\frac{\partial f(-1,2)}{\partial y}} = -2$$

Therefore, the equation of the tangent line h to the level curve L at $(-1, 2)$ is

$$h(x) = y_0 + g'(x_0)(x - x_0) = 2 + (-2)(x - (-1)) = -2x$$

3. Finally, in this case, from

$$\frac{\partial f(x,y)}{\partial x} = y$$
$$\frac{\partial f(x,y)}{\partial y} = x$$

it follows that f is continuously differentiable, and that $\frac{\partial f(1,2)}{\partial y} = 1 \neq 0$. Hence by the implicit function theorem, a continuously differentiable function $y = g(x)$ whose graph describes L in a vicinity of $(1, 2)$ exists, and

$$g'(1) = -\frac{\frac{\partial f(1,2)}{\partial x}}{\frac{\partial f(1,2)}{\partial y}} = -2$$

Therefore, the equation of the tangent line h to the level curve L at $(1, 2)$ is

$$h(x) = y_0 + g'(x_0)(x - x_0) = 2 + (-2)(x - 1) = -2x + 4$$

Problem 5

Calculate the Hessian matrix of each of the following functions.

(a) $f : \mathbb{R}^2 \to \mathbb{R}, (x, y) \mapsto xe^{xy}$

Solution: (Sketch only) We obtain

$$\nabla f(x, y) = \begin{pmatrix} e^{xy} + xye^{xy}, & x^2e^{xy} \end{pmatrix}$$
$$H(x, y) = e^{xy} \begin{bmatrix} 2y + xy^2 & 2x + x^2y \\ 2x + x^2y & x^3 \end{bmatrix}$$

(b) $f : \mathbb{R}^2_{++} \to \mathbb{R}, (x, y) \mapsto \ln(xy + y^2 + x^2)$

Solution: (Sketch only) We obtain

$$\nabla f(x, y) = \left(\frac{y + 2x}{xy + y^2 + x^2}, \frac{x + 2y}{xy + y^2 + x^2} \right)$$
$$H(x, y) = \frac{1}{(xy + y^2 + x^2)^2} \begin{bmatrix} y^2 - 2xy - 2x^2 & -x^2 - y^2 - 4xy \\ -x^2 - y^2 - 4xy & x^2 - 2y^2 - 2xy \end{bmatrix}$$

(c) **(Harder; Extra Practice Problem)** $f : \{(x, y) \in \mathbb{R}^2_+ : xy \leq 3\} \to \mathbb{R}, (x, y) \mapsto \sqrt{3x - x^2y}$

Solution: (Sketch only) We obtain

$$\nabla f(x, y) = \left(\frac{3 - 2xy}{2\sqrt{3x - x^2y}}, \frac{-x^2}{2\sqrt{3x - x^2y}} \right)$$
$$H(x, y) = \frac{1}{4\,(3x - x^2y)^{\frac{3}{2}}} \begin{bmatrix} -9 & -9x^2 + 2x^3y \\ -9x^2 + 2x^3y & -x^4 \end{bmatrix}$$

Problem 6 (Extra Practice Problem)

Suppose that it follows from the implicit function theorem that, in a neighborhood of (x_0, y_0, z_0), z is implicitly defined by the equation $x^2 + y^2 + z^2 = a^2$ as a function of x and y. Compute the total derivative $Dz(x_0, y_0)$.

Solution: If we let $f : \mathbb{R}^3 \to \mathbb{R}, (x, y, z) \mapsto x^2 + y^2 + z^2 - a^2$ then the equation above is equivalent to $f(x, y, z) = 0$. We have

$$\frac{\partial f(x,y,z)}{\partial x} = 2x$$
$$\frac{\partial f(x,y,z)}{\partial y} = 2y$$
$$\frac{\partial f(x,y,z)}{\partial z} = 2z$$

(Since all partial derivatives of f are continuous, f is continuously differentiable.) Since the implicit function theorem applied, $\frac{\partial f(x_0,y_0,z_0)}{\partial z} = 2z_0 \neq 0$. By the implicit function theorem,

$$\frac{\partial z(x_0,y_0)}{\partial x} = -\frac{\frac{\partial f(x_0,y_0,z_0)}{\partial x}}{\frac{\partial f(x_0,y_0,z_0)}{\partial z}} = -\frac{2x_0}{2z_0} = -\frac{x_0}{z_0}$$
$$\frac{\partial z(x_0,y_0)}{\partial y} = -\frac{\frac{\partial f(x_0,y_0,z_0)}{\partial y}}{\frac{\partial f(x_0,y_0,z_0)}{\partial z}} = -\frac{2y_0}{2z_0} = -\frac{y_0}{z_0}$$

Therefore, $Dz(x_0, y_0) = \nabla z(x_0, y_0) = \left(-\frac{x_0}{z_0}, -\frac{y_0}{z_0}\right)$.

(Alternatively, from $Df(x_0, y_0, z_0) \cdot (dx, dy, dz) = 0$ we obtain

$$(2x_0, 2y_0, 2z_0) \cdot (dx, dy, dz) = 2x_0 dx + 2y_0 dy + 2z_0 dz = 0$$

or

$$dz = -\frac{2x_0}{2z_0}dx - \frac{2y_0}{2z_0}dy = -\frac{x_0}{z_0}dx - \frac{y_0}{z_0}dy,$$

and we can see that $Dz(x_0, y_0) = \left(-\frac{x_0}{z_0}, -\frac{y_0}{z_0}\right)$.)